

SENSEI: A Simple, Elegant, and Nimble Subtitle Editing Interface

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Abstract

We present SENSEI (Simple, Elegant, and Nimble Subtitle Editing Interface), a fully open-source, web-based human-in-the-loop tool for the post-editing of automatically generated subtitles. SENSEI is designed to support operators in correcting and validating subtitles produced by an ASR-MT cascade within an ergonomic workspace. By providing parallel management of source and target tracks, the interface enables genuinely bilingual post-editing. At the same time, it facilitates compliance with industry-standard formatting constraints in terms of temporal synchronization, reading speed, line length, and lines per block. The interface integrates a video preview module, a block-based text editor, and a multimodal timeline with waveform visualization, all synchronized in real time. The underlying automatic subtitling system, briefly described in this paper, is a two-stage ASR-MT cascade built exclusively from freely available components, requiring no proprietary external APIs. Both the interface and the underlying subtitling pipeline are publicly released under the Apache-2.0 license, making the entire end-to-end system immediately usable for research and commercial purposes alike.

Keywords

Subtitle editing, human-in-the-loop, automatic subtitling, ASR-MT cascade

1. Introduction

The growing demand for high-quality subtitles across broadcast, streaming, and online video platforms has driven significant research interest in automatic subtitling systems. Unlike standard speech translation, subtitling imposes a set of spatiotemporal constraints on the output, including reading speed, line length, and the number of lines per subtitle block, that go beyond mere translation quality and directly affect the viewer’s cognitive experience. While fully automatic approaches have made remarkable progress in recent years [1, 2, 3], the gap between automatically generated subtitles and broadcast-ready output still makes human post-editing a practical necessity in many professional scenarios.

In this paper, we present an integrated subtitling environment consisting of three components. The first is an ASR-MT cascade for the automatic generation of subtitles from audiovisual documents, previously described in detail in [4]. The cascade is built from freely available models and is capable of producing subtitles in multiple target languages while enforcing compliance with standard formatting guidelines. The second and primary contribution of this work is SENSEI (Simple, Elegant, and Nimble Subtitle Editing Interface), a web-based editing interface that enables human operators to inspect, correct, and validate the automatically generated output within a single, ergonomically designed workspace. Finally, the third component is a persistent database, where settings, users and projects are stored.

SENSEI is conceived as a Human-in-the-Loop (HITL) tool that places the operator at the center of the post-editing workflow. Rather than requiring manual interaction with raw SRT files or generic text editors, SENSEI provides a purpose-built environment in which textual content, timestamps, compliance

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metrics, and video playback are continuously synchronized. SENSEI stands out from traditional open-source editors by combining three core capabilities that are typically absent in general-purpose tools: 1) compliance metrics are treated as first-class, per-block signals with user-editable thresholds and live violation flagging; 2) the editing operators are tailored to the characteristic error patterns of the upstream ASR-MT cascade; and 3) the source and target tracks, representing intra-lingual captions and cross-lingual subtitles respectively, are kept in reactive synchronization across all panels for genuinely bilingual post-editing, enabling operators to visually compare and edit both cascade stages simultaneously. Operators can thus perform all necessary corrections, from textual edits to temporal boundary adjustments, with immediate visual and auditory feedback, minimizing both error rates and editing time.

Figure 1 provides an overview of the overall SENSEI architecture: a Vue-based frontend that supports human-in-the-loop subtitle editing, a management backend responsible for user and project administration, and the automatic subtitling backend implementing the ASR-MT pipeline (see Section 4 and Figure 5 for more details). Together, these components support the complete workflow from audiovisual input and automatic subtitle generation to bilingual post-editing and final SRT export.

A distinctive feature of our contribution is its full commitment to open tooling: both the subtitling pipeline and the SENSEI front-end are publicly released on GitHub under the Apache-2.0 license, together with all the component models, making the complete system the first entirely self-hostable ecosystem freely usable for research and commercial applications without restrictions or reliance on commercial third-party services.

The remainder of this paper is organized as follows. Section 2 provides an overview of the recent tools and literature that defines the scope of our work. Section 3 describes the SENSEI interface in detail. Section 4 summarizes the underlying automatic subtitling system. Section 5 discusses the code release and licensing terms.

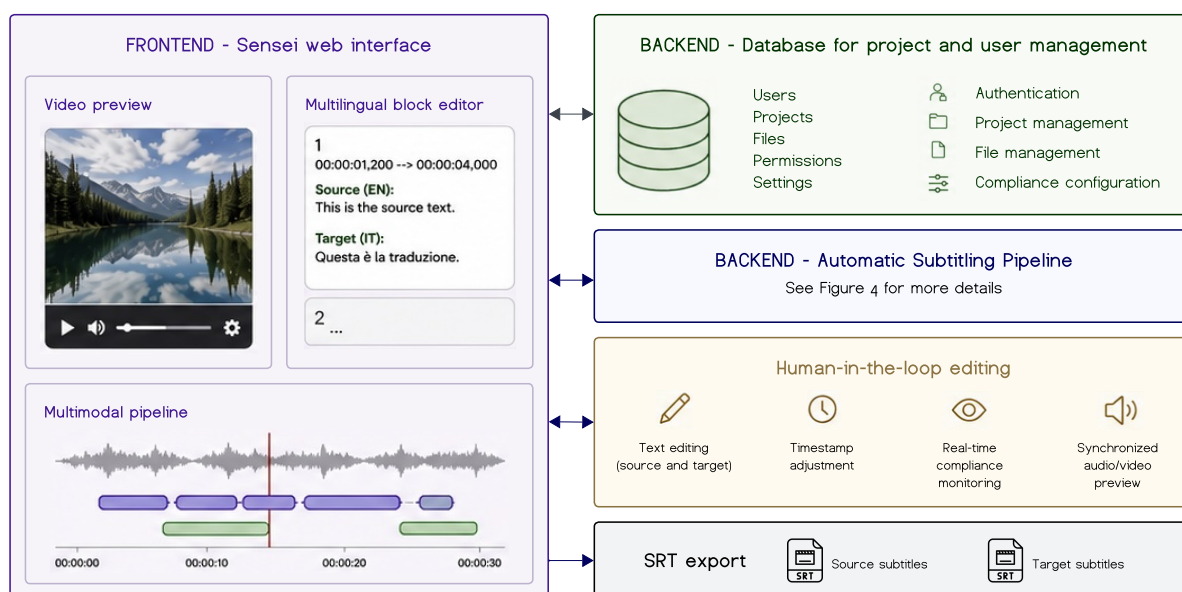


Figure 1: Overall architecture of the SENSEI ecosystem.

2. Related Work

Commercial subtitling and editing platforms. Several commercial platforms offer integrated pipelines for automatic subtitle generation and subsequent human post-editing. Adobe Premiere

Pro¹ is a professional video editing suite that incorporates automatic speech-to-text transcription and subtitle generation, tightly integrated with a full-featured non-linear video editing environment. Matesub² is a web-based platform specifically designed for professional subtitlers, combining automatic transcription and translation with a dedicated subtitle editor that enforces compliance with industry-standard formatting constraints. HappyScribe³ offers an online subtitle editor built around automatic transcription and translation services, targeting both individual content creators and media companies. Amara⁴ is a collaborative subtitling platform that supports both manual and semi-automatic subtitle creation, with particular attention to accessibility and community-driven workflows. While these tools, along with many others not mentioned here, provide polished user experiences, they operate under proprietary licenses and, beyond a typically limited free plan, require a paid subscription, making them unsuitable for unrestricted research use or large-scale academic deployment.

Open-source subtitle editors. Several free and open-source editors are widely used for subtitle creation and post-editing. Aegisub⁵ and Gaupol⁶ provide mature timing and styling environments but no built-in automatic generation. Subtitle Edit⁷ is the most feature-complete solution: it integrates Whisper-based transcription, waveform/spectrogram visualization, and post-processing tools for line breaking and timing adjustment, and supports translation through external services. However, as a desktop-oriented application, it is designed for use on a single workstation, making multi-user collaboration on the same project less straightforward than in web-based platforms. It relies on locally installed AI models for transcription and, when applicable, translation. Translation can also be performed through external APIs (e.g., Google Translate or DeepL), although this requires access to third-party services, which are typically subject to usage fees.

Automatic subtitling systems. Since 2023, IWSLT has organized an annual shared task on automatic subtitling⁸ [5], providing a standardized evaluation framework with the aim of accelerating progress in the field. The dominant and consistently best-performing approach is the ASR-MT cascade, in which speech recognition and machine translation are applied sequentially, complemented by post-processing strategies to enforce spatiotemporal compliance, including constraints on reading speed, line length, and lines per subtitle block, with the goal of reducing the cognitive load imposed on viewers. Among the strongest systems to emerge from these evaluations are those developed at AppTek [6] and Huawei Translation Services Center [7, 8], which represent the state of the art in terms of subtitle quality across multiple language pairs and domains. The former is integrated into a commercial service, while the latter was developed for research purposes only; in both cases, neither the systems nor their components are publicly available.

Positioning of SENSEI. To the best of our knowledge, SENSEI is the only complete subtitling environment that covers both automatic subtitle generation and human post-editing while being released in its entirety under a permissive license and freely usable without restrictions, including for commercial purposes. As a fully web-based system, it requires no local installation and can be accessed from any workstation through a browser, regardless of the operating system. This makes it particularly suitable for distributed teams and shared deployments in which installing and maintaining desktop applications would be impractical. Moreover, unlike traditional editors that are primarily designed around a single subtitle track, SENSEI provides a natively bilingual workspace in which source captions and target translations remain interactively synchronized across all panels, enabling

¹<https://www.adobe.com/it/products/premiere.html>

²<https://matesub.com>

³<https://www.happyscribe.com/tools/online-subtitle-editor>

⁴<https://amara.org>

⁵<https://aegisub.org/>

⁶<https://otsaloma.io/gaupol/>

⁷<https://www.nikse.dk/subtitleedit>

⁸<https://iwslt.org/2026/subtitling>

simultaneous comparative post-editing. While other editors may optionally interface with external AI models, SENSEI directly integrates the ASR-MT cascade into its self-hostable pipeline. This integration provides out-of-the-box support for hundreds of source and target languages within a fully autonomous ecosystem that does not depend on third-party orchestration or paid APIs.

Figure 2 further clarifies this positioning through a feature-based comparison with representative open-source subtitle editors. Although the considered tools generally provide the core operations required for subtitle editing, SENSEI brings together automatic subtitle generation, mouse- and keyboard-based timing adjustment, configurable compliance constraints, SRT import, and a fully resizable web interface within a single environment. Most importantly, it natively manages synchronized source and target tracks, supporting bilingual post-editing without requiring separate projects or workspaces.

Functions	SENSEI	MateSub	HAPPYSCRIBE	Aegisub	Subtitle Edit
Automatic subtitle generation	☒	☒	☒	☐	☒
Merge	☒	☒	☒	☒	☒
Add	☒	☒	☒	☒	☒
Delete	☒	☒	☒	☒	☒
Duplicate	☒	☐	☐	☒	☐
Edit text	☒	☒	☒	☒	☒
Edit timestamp (keyboard)	☒	☒	☒	☒	☒
Edit timestamp (mouse)	☒	☒	☒	☐	☒
Fullscreen video	☒	☐	☒	☐	☒
Resizable interface	☒	Just open/close sidebar	Only sidebar	☒	☒
Multi-language	☒	☐	Separate environments	☐	☐
Load SRT file	☒	☒	☒	☒	☒
Playback of single subtitle	☒	Only from vertical sidebar	☐	☒	Only from vertical sidebar

Figure 2: Comparison between SENSEI and open-source alternatives.

3. User Interface - SENSEI

The front-end module of the architecture (left side block in Figure 1) consists of the SENSEI interface. It represents the presentation and human-machine interaction layer that enables the human operator to visualize, manipulate, and validate the subtitles automatically generated by the subtitling pipeline described in Section 4.

While the preliminary and initial configuration processes, such as user authentication, session management, source video upload, and the configuration of input and output language pairs, are briefly discussed in Section 3.5, the main SENSEI workspace is specifically focused on the analytical visualization and editing environment for synchronized textual streams.

Within the context of *Computer-Assisted Translation and Transcription* systems, SENSEI is conceived as a *Human-in-the-Loop* (HITL) tool (right side, third block in Figure 1). Its primary objective is the optimization of post-editing workflows by providing the user with an ergonomic infrastructure for correcting potential textual inaccuracies generated by the model.

At the same time, the interface enables the verification and validation of segment compliance with the metric standards imposed by the subtitling industry. Among these critical parameters, the system allows the monitoring and correction of:

- The maximum number of lines allowed within a single subtitle block;
- The linear length (in number of characters) of each line, in order to prevent visual overflow;
- The reading speed, dynamically calculated in characters per second (CPS) according to the following formula:

$$CPS = \frac{C}{t_{out} - t_{in}}$$

where C represents the total number of characters within the block and $(t_{out} - t_{in})$ its temporal duration;

- The absolute on-screen duration of each audiovisual segment.

All threshold values associated with the aforementioned standards can be modified by the user. SENSEI evaluates these metrics per block and continuously, re-computing them after every textual or temporal edit. Blocks that violate the active thresholds are flagged in place with a visual warning indicator (see Figure 3), so that the operator is always aware of, and can immediately resolve, the current compliance state.

To minimize the operator’s cognitive load and ensure smooth navigation, the workspace is structured into four interconnected and real-time synchronized macro-functional areas: the top management bar, the video preview monitor, the block-based text editor, and the multimodal timeline.

A fundamental engineering principle of the front-end is its layout flexibility: all three primary operational work areas (video preview, text editor, and timeline) are dynamically resizable through direct mouse interaction. By dragging the lateral or upper dividers, the user can continuously reshape the geometric extension of the panels, adapting the interface to the priorities of the current editing phase.

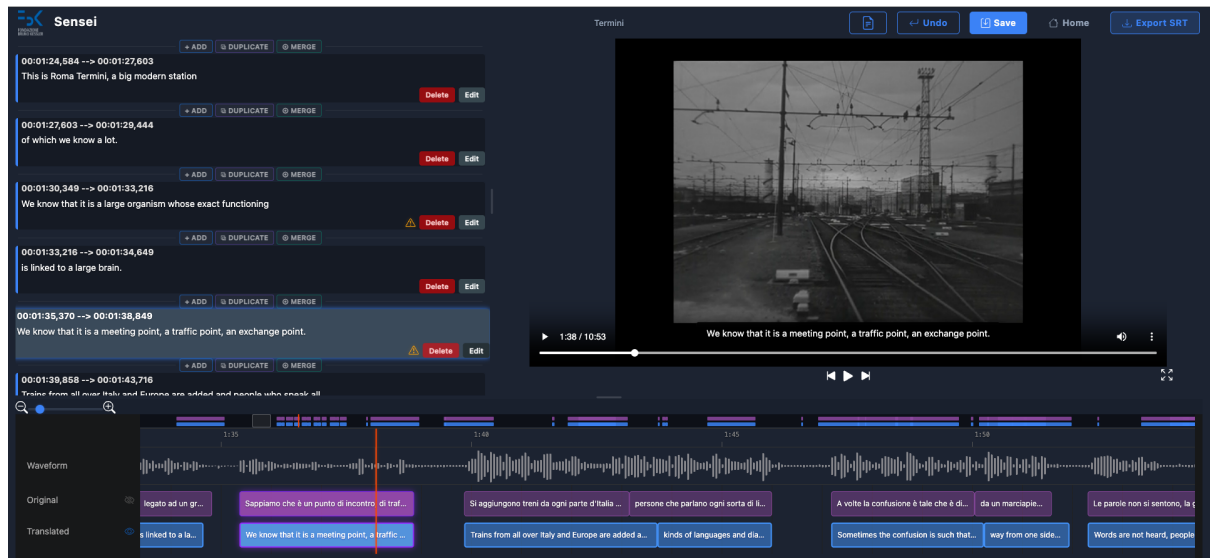


Figure 3: SENSEI editing interface: top bar with system functions (top), video preview module (right), block-based text editor (left), and multimodal timeline with audio waveform visualization (bottom).

3.1. Top Bar: System Functions

The upper region of the interface hosts the application header, a centralized command bar that handles project state management and data persistence within the database (as depicted in the top-right block of Figure 1). The components are distributed along the horizontal axis according to a precise functional hierarchy:

Brand Identity and In-Place Project Title Editing: On the far left, the system logo (Sensei) is displayed. At the center of the bar, the title of the current project is rendered. To avoid the introduction of contextual menus or intermediate modal windows, the interface implements an in-place editing mechanism: the user can modify the project name directly by clicking on the text string, thereby optimizing interaction time.

Undo Function: The front-end integrates an error-tolerance mechanism based on an *Undo* command. This tool manages a retroactive recovery buffer capable of storing the last 50 operations performed

by the user (e.g., textual modifications, temporal shifts, deletions) prior to definitive saving. This ensures protection against accidental changes typically occurring during intensive editing sessions.

Progress Saving and Project Closure: The *Save* button activates the synchronous serialization of all structural subtitle modifications to the database. Conversely, the *Home* button allows the user to exit the editing environment and return to the general dashboard of saved projects. To prevent unintentional loss of progress, the system intercepts the exit action: if unsaved changes are detected, the front-end activates a conditional control routine requiring explicit user confirmation before redirecting to the *My Projects* section.

Subtitle File Export: The *Export* (or *Export SRT*) command performs the immediate download of subtitle files, preserving the exact formatting and alignment state present at the moment the button is pressed. The system simultaneously generates two standard SRT (*SubRip Subtitle*) files, corresponding respectively to the source and target languages. In order to guarantee file traceability and organization within local file systems, the front-end applies an automatic naming pipeline structured according to the following formal grammars:

projectName_sourceLanguageCode_date

projectName_targetLanguageCode_date

3.2. Video Preview Monitor: Real-Time Visual Validation

In the right section of the interface, the *Video Player* represents the module responsible for immediate visual feedback. Its function is not merely reproductive; rather, it serves as a control instrument for visual synchronization and the aesthetic inspection of overlaid text.

The monitor includes a lower control bar equipped with a set of standard commands:

- Playback state buttons (*Play/Pause*);
- Full-screen mode activation (*Full Screen*) for high-resolution inspection;
- Fast navigation buttons configured to instantly move the playback head to the exact beginning or end of the currently loaded video sequence.

3.3. Text Editor: Block Management

The left region of the interface hosts the analytical subtitle segment editor. The front-end data architecture maps the textual stream as an ordered sequence of discrete blocks. Each block explicitly displays the linguistic content of the subtitle (distributed over one or more lines according to spatial formatting) together with its associated temporal metadata, namely the numerical input (*In*) and output (*Out*) timestamps expressed in the standard high-precision format `hours:minutes:seconds,milliseconds`.

Bilingualism and Bidirectional Workflows: The interface implements advanced multilingual management designed for bidirectional workflows (e.g., Italian and English). Whereas most open-source editors are organized around a single subtitle track, SENSEI treats source and target as first-class parallel tracks, making it equally suited to intra-lingual caption correction and cross-lingual translation post-editing. Through a tab-based interface positioned at the top (*Original* and *Translated*), the operator can instantly switch between the visualization and editing of the source-language track and the target-language track, ensuring continuous comparative control over both stages of the generation pipeline.

Structural Operators on Segments: The set of editing operators is deliberately shaped around the recurrent error patterns of the ASR-MT subtitling cascade (Section 4), so that the most frequent corrections map onto single, dedicated commands:

- **Edit:** Enables text fields and numerical timing values for precise keyboard-based modifications;
- **Delete:** Removes the logical segment from the chronological stream;
- **Add:** Manually inserts a new empty segment between adjacent blocks;
- **Duplicate:** Clones the current block, inheriting its textual content. This is useful for manually segmenting subtitles that violate the character-per-line or line-count active thresholds. By duplicating the non-compliant block and deleting the redundant text from each instance, the operator can distribute the text via subtraction without re-typing, thereby streamlining the layout adjustment workflow and eliminating the risk of introducing typos;
- **Merge:** Performs the logical and temporal fusion of two contiguous blocks. This operator is crucial in NLP editing to correct over-segmentation errors induced by the audio segmentation algorithms (Section 4.2).

Navigational Consistency and Ergonomic Feedback: To facilitate the rapid identification of typographical errors or synchronization mismatches during playback, SENSEI adopts an automatic highlighting system. The textual block corresponding to the video frame currently being played in the monitor is chromatically highlighted by the front-end. Navigation across the entire textual corpus, which may extend to thousands of lines in long-form videos, is supported by a high-fluidity vertical scrolling bar implementing predictive position tracking.

3.4. Multimodal Timeline: Spatiotemporal Synchronization

The lower region of the interface hosts the Multimodal Timeline, architecturally the most relevant element for the management of cross-modal information (audio, video, and text). The timeline displays subtitle blocks along a temporal axis regulated by a ruler positioned at the top of the section, with time explicitly expressed in minutes and seconds.

Waveform Extraction and Visualization: The front-end renders the waveform of the original audio track. This graphical representation enables the user to visually identify variations in signal amplitude, locating speech activity and pauses/silences. The operator can therefore position subtitles on the basis of an objective morphological reference, reducing the need for continuous listening and re-listening cycles.

Spatial Manipulation of Temporal Boundaries: Overcoming the operational limitations of manual numerical timestamp insertion, which is inherently prone to human error, SENSEI represents subtitles as interactive two-dimensional geometric boxes positioned along the timeline. The user can adjust subtitle duration and timing by dragging the lateral borders (interval resizing) or translating the entire block along the horizontal axis (*drag-and-drop* operations). This visual interaction paradigm prevents the occurrence of unintended temporal overlapping between consecutive subtitles.

Reactive State Synchronization: The original transcription and target translation tracks are displayed in parallel along the vertical axis of the timeline. Through dedicated eye-shaped visibility icons, the operator selects the specific track on which to operate. This selection is not isolated; rather, it triggers a dynamic and synchronous update of the entire application state: it instantly modifies the textual content visible within the block editor (Section 3.3) and updates the textual overlay displayed within the video preview module (Section 3.2), thereby preserving system-wide consistency. This track-level reactivity is what makes the bilingual workflow practical: switching the operating track is a single action that consistently propagates to the editor, the timeline, and the video overlay, rather than a manual realignment across disjoint views.

Tracking and Frame-Level Inspection Devices: The current video position is mapped onto the timeline through a red vertical playback head (*playhead*). The subtitle active at that specific instant is indicated through a high-visibility contour around the corresponding timeline box. To enable precise timing adjustments, the interface provides zoom control tools. Composed of magnifying-glass buttons (+ and −) and a continuous linear slider, these controls allow the expansion or contraction of the timeline temporal scale up to single-frame resolution, maximizing subtitle boundary placement accuracy.

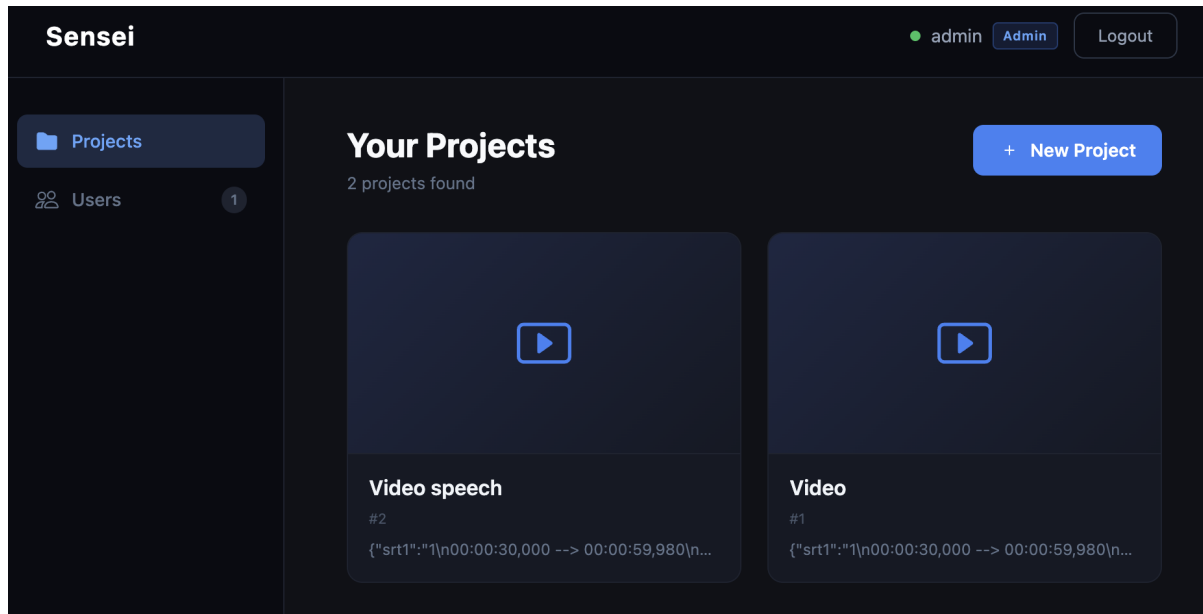


Figure 4: SENSEI admin interface where users and projects can be created and managed.

3.5. Dashboard and Project Creation

The operational cycle within the ecosystem begins with an authentication and configuration phase designed to ensure user data security, persistence, and isolation. Upon logging in, the operator is automatically redirected to the main management dashboard, named *My Projects* (see Figure 4). This centralized interface serves as the control hub for the user's repository; from here, it is possible both to browse and monitor the list of pre-existing projects – enabling operators to quickly resume previous editing sessions – and to initiate the creation of a new project. The initialization of a new instance occurs by clicking the *Create New Project* button, which activates a structured configuration modal window designed to set up the parameters of the processing pipeline. Within this module, the operator is required to enter the project's identifying metadata, such as the project name, and upload the multimedia video file from the local file system.

In order to maximize the flexibility of the system with respect to different industrial scenarios associated with audiovisual material, the form integrates an interactive mode selector (*slider*). Through this component, the user can determine the subtitle processing strategy by selecting between two distinct approaches:

- **Automatic Generation:** Activates the pipeline described in Section 4, which requires the preliminary specification of the source language of the video's audio stream and the corresponding target translation language;
- **External Asset Import:** Supports the direct upload of pre-existing SRT (*SubRip Subtitle*) files from the computer's local storage. This mode bypasses the automatic transcription and translation models, configuring the SENSEI environment as a pure editing and alignment tool for already-consolidated subtitle texts. This setup is particularly valuable for performing targeted, late-stage

revisions on existing materials without forcing the user to regenerate the entire subtitle track from scratch.

4. Automatic Subtitling System

The automatic subtitling system whose output is handled by the editing interface described in this paper is a two-stage ASR-MT cascade built exclusively from freely available components, all usable without restrictions including for commercial purposes [4]. Its overall architecture is illustrated in Figure 5.

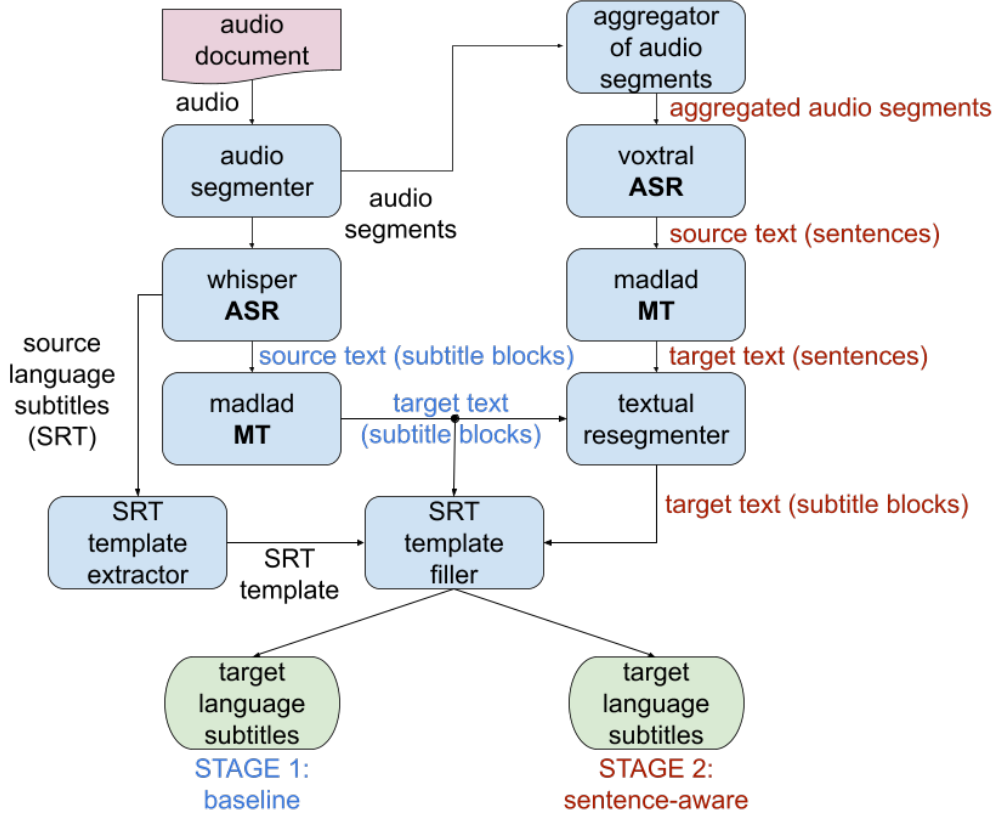


Figure 5: The two-stage subtitling architecture.

4.1. Models

The system draws on the following core components. SpeechBrain VAD (SB VAD; [9]) is a voice activity detection model that segments the input audio into speech regions, filtering out silence and non-speech portions. Whisper large-v3 [10], a 1.55B-parameter multilingual ASR model trained on 680,000 hours of labeled audio, is used for transcription; crucially, it natively supports SRT output generation with token-level timestamp alignment. VOXTRAL [11], a 3B-parameter ASR model from Mistral AI, features a 32k-token context window enabling high-quality transcription of long-form audio, but it is unable to produce time-aligned output directly. Finally, MADLAD-400-10B-MT [12] is a 10.7B-parameter multilingual MT model supporting over 400 languages, used for all translation steps with 4-beam decoding and no length penalty.

4.2. Stage 1: Baseline Subtitling

The first stage follows a conventional subtitle-level ASR-MT pipeline. The audio extracted from the input audiovisual document is first segmented by SB VAD. Each segment is then transcribed by Whisper, which produces an SRT file with subtitle-level blocks and associated timestamps. The text of each

subtitle block is translated by MADLAD, and the resulting translation replaces the original transcript in the SRT file. The output of this stage constitutes the baseline subtitling system.

4.3. Stage 2: Sentence-aware Refinement

The second stage builds on the output of Stage 1 with a refinement step motivated by a key observation: subtitle-level segments are often too short and fragmentary to be optimal input for either ASR or MT, whereas longer, sentence-like units consistently lead to better transcription and translation quality. Adjacent SB VAD segments are therefore aggregated into longer audio units, subject to two heuristic constraints: a maximum gap of 10 seconds between consecutive segments, and a maximum resulting duration of 600 seconds. These longer units are transcribed by VOXTRAL, which unlike Whisper-based models can effectively exploit the extended acoustic and linguistic context they provide, yielding lower word error rates than Whisper on the same material. The resulting source text is then split into sentence units using strong punctuation cues, or by enforcing a maximum length when punctuation is absent, and translated sentence by sentence by MADLAD. Sentence-level translation is more natural for the MT model, which is trained on well-formed sentences rather than subtitle fragments, thereby yielding more fluent and adequate output. The sentence-level translations are finally word-level re-aligned to the subtitles produced in Stage 1 using mweralign [13], and inserted into the original SRT template. This design preserves subtitle synchronization while leveraging broader linguistic context to improve textual quality.

4.4. Post-processing

Both stages are followed by a set of post-processing steps enforcing subtitle compliance. If a subtitle’s reading speed exceeds the predefined threshold (21 cps for non-CJK languages, 9 cps for Chinese, 4 cps for Japanese), its end timestamp is extended to the minimum value satisfying the constraint, up to the start time of the following subtitle. Subtitles exceeding the maximum line length (42/16/13 characters for non-CJK, Chinese, and Japanese respectively) are split into lines of balanced length, prioritizing punctuation boundaries; if more than two lines would result, the subtitle is further divided into multiple blocks with proportionally assigned timestamps. Finally, a hallucination filter removes consecutive repetitions of n-grams (for n from 4 down to 1, applied iteratively), mitigating a known failure mode of both ASR and MT models where the output loops over the same fragment rather than producing meaningful content.

4.5. Performance

Evaluation on the IWSLT 2026 Subtitling shared task introduced in Section 2, conducted across three domains (ITV entertainment, Asharq-Bloomberg financial news, and YODAS YouTube recordings) and up to four target languages per domain chosen from a pool of five (Arabic, Chinese, German, Japanese, Spanish), confirms the effectiveness of the two-stage design. The primary system consistently outperforms the single-stage baseline across all conditions, with the largest gains observed in translation quality metrics (BLEU, ChrF, BLEURT). These improvements are attributable to two complementary factors: more accurate transcription from longer audio segments, and higher translation quality from sentence-level MT input. On the ITV test2023 progressive benchmark, used across successive IWSLT editions to track progress over time, the system proves competitive with AppTek’s commercial solution on German and Spanish while outperforming all previous FBK submissions by a wide margin in terms of both translation quality and overall subtitle quality (SubER). For a full description of the system architecture, model selection experiments, and detailed results across all domains and language pairs, the reader is referred to [4].

5. Code Release and Licenses

The front-end module (the SENSEI interface, Section 3), dedicated to the visualization and post-editing of subtitles generated by the automatic subtitling system (Section 4), has been developed using HTML/CSS/VueJS and released under the Apache-2.0 license. The automatic subtitling module is accessible via the Python/FastAPI paradigm; the code is released under the Apache-2.0 license. The underlying models, namely:

- SpeechBrain VAD
- openai/whisper-large-v3 ASR model
- mistralai/Voxtral-Mini-3B-2507 ASR model
- google/madlad400-10b-mt MT model

are likewise distributed under the Apache-2.0 license by their respective authors. The code of both modules is publicly available on two Github repositories:

- <https://github.com/hlt-mt/sensei>
- <https://github.com/hlt-mt/subtitler>

The first repository contains the software of the SENSEI interface for editing subtitles, and of the layer managing persistent project and user information. The second repository offers the code to run the subtitling pipeline, based on two docker images, three archives with models and the software to orchestrate the webservice requests.

6. Conclusion

We have presented SENSEI, a web-based human-in-the-loop environment for the post-editing of automatically generated subtitles. SENSEI integrates a purpose-built editing interface with an ASR-MT cascade capable of producing subtitles in hundreds of languages, covering the full subtitling workflow from raw audiovisual input to broadcast-ready output. A key distinguishing feature of the system is its commitment to openness: all components, including the front-end interface, the subtitling pipeline, and the underlying models, are publicly released under permissive licenses, making SENSEI immediately usable for both research and commercial applications without restrictions.

Beyond its open licensing, SENSEI’s interface is purpose-built around the post-editing loop itself, featuring live per-block compliance, operators tailored to the cascade’s error patterns, and reactive bilingual track synchronization designed to streamline both intra-lingual captioning and cross-lingual translation workflows. Furthermore, by supporting the direct import of pre-existing resources, the environment doubles as a lightweight tool for targeted, late-stage subtitle revisions that bypasses the automated pipeline entirely. To the best of our knowledge, this makes SENSEI the only complete subtitling environment released in its entirety under a permissive license.

Future work will focus on a systematic evaluation of the SENSEI interface in realistic post-editing scenarios. We plan to conduct user studies with professional and semi-professional subtitle operators at partner sites, comparing SENSEI with established subtitle editors on shared editing tasks. The evaluation will assess both objective measures, such as task completion time, editing accuracy, number and type of corrections, and compliance violations, as well as subjective measures, including perceived usability, cognitive load, and user satisfaction. This will allow us to empirically validate the effectiveness of the interface and determine whether its bilingual workflow, synchronized views, and integrated compliance monitoring provide measurable benefits over existing tools.

On the automatic subtitling side, we plan to investigate more sophisticated cross-modal alignment strategies to further improve temporal synchronization, addressing current limitations in complex acoustic and multi-speaker environments.

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Declaration on Generative AI

During the writing of this article, the authors used the `claude.ai` chat (free plan) in order to re-write in English some paragraphs originally drafted in Italian. After using the tool, the authors reviewed and edited the content as needed and take full responsibility for the publication’s content.

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